

Ganesh Gupta

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EXPERIENCE

Oracle | *Software Engineer, AI Platform*

Jun 2025 – Present

- Provisioned and operated multi-cloud GPU clusters (AWS, GCP, CoreWeave) with NVIDIA/CUDA hardware and InfiniBand/RoCE networking, supporting long-running training jobs and high-availability inference on shared fleet capacity.
- Wrote infrastructure/systems-level code in Go, Rust, and C++ for scheduling, resource allocation, and fault-tolerant placement logic under GPU-scarcity constraints.
- Built observability and autoscaling for vLLM inference serving via Prometheus/Grafana; ran distributed training workloads via Slurm-based HPC scheduling, sustaining fleet utilization through node loss and provider hiccups.
- Built a multi-agent GenAI system via OpenAI Agents SDK and MCP tool-use, integrating ML models into production workflows handling millions of daily transactions at 99%+ uptime.
- Designed and optimized agentic AI workflows across multiple LLMs (GPT, Claude, Llama, Gemini), selecting and tuning models per use case for production quality and cost.

QureAI | *ML Infrastructure Engineer*

Mar 2026 – Present

- Extended and optimized inference framework internals (vLLM-class serving) with kernel-level development (CUDA/Triton), distributed KV cache, and speculative decoding to improve LLM serving performance.
- Ran PyTorch model training/fine-tuning and model compression/quantization on multi-GPU clusters for efficient production inference.
- Cut p99 inference latency 420ms to 140ms across 50K+ daily requests by migrating synchronous model serving to async vLLM on GCP Cloud Run with autoscaling.

Nomura Research Institute Financial Technologies | *Software Engineer*

Sep 2019 – Aug 2023

- Built and operated 25+ production microservices in Java, C++, and SQL-backed distributed systems for 1M+ users, sustaining 99.9% uptime over 3 years on AWS (EC2, RDS, SQS).
- Cut nightly financial data aggregation blocking downstream settlement from 8 minutes to 90 seconds by redesigning SQL with indexed views and batch partitioning.
- Standardized failure-handling contracts across 25+ financial services, reducing production escalations 60% in one quarter via consistent retry, timeout, and propagation semantics.

PROJECTS

Watch-your-LLM

github.com/ganeshhgupta/watch-your-llm

- Production LLM eval and observability platform with rubric-based scoring, drift detection, and latency tracking across model versions; FastAPI backend, React/TypeScript dashboard, PostgreSQL.

RootCause AI

github.com/ganeshhgupta/rootcause-ai

- LangGraph multi-agent system diagnosing production incidents via tool-use chains; routes alerts to specialized sub-agents and synthesizes root cause reports.

Distributed KV Store / VPC Control Plane Simulator

github.com/ganeshhgupta/distributed-kv-store

- Raft-consensus distributed KV store handling concurrent writes with linearizable reads and log compaction; companion VPC control plane simulator modeling subnet routing and security group enforcement.

EDUCATION

University of Texas at Arlington | *M.S. Computer Science, GPA 4.0/4.0*

Aug 2023 – May 2025

MCKV Institute of Engineering | *B.Tech Computer Science*

Jul 2016 – Jul 2020

SKILLS

Languages: Python, Java, C++, Go, Rust, TypeScript, SQL

AI/ML & GenAI: LLMs, RAG, Fine-Tuning, PyTorch, TensorFlow, Hugging Face, LangChain, LangGraph, MCP

GPU & HPC: CUDA, Triton, NVIDIA GPU Clusters, InfiniBand/RoCE, Multi-GPU Training, Model Compression/Quantization, Slurm, vLLM

MLOps & Distributed Systems: Distributed Systems, Kubernetes, Docker, Prometheus/Grafana, PostgreSQL, Redis, BigQuery

Cloud & DevOps: AWS (EC2, S3, SageMaker, Lambda), GCP (Cloud Run, Vertex AI), CoreWeave, CI/CD

CERTIFICATIONS

AWS Certified Developer Associate | Google Cloud Professional Machine Learning Engineer